UAV Acronym Glossary

Course: U21MI504 – Unmanned Aerial Vehicle Technology · all units
Unit column shows where the term is used most. Terms follow the course reader (Chapters 1–5).

Acronym	Meaning	Unit
ADC	Analog-to-digital converter	IV
ADS-B	Automatic Dependent Surveillance–Broadcast	I, V
AGL	Above ground level	III, V
AltHold	Altitude hold flight mode	V
AR	Aspect ratio (b^2/S)	II
ARC	Air risk class (SORA)	V
ATC	Air traffic control	V
BEC	Battery eliminator circuit (regulated supply for avionics)	IV
BIM	Building information modeling	III
BLDC	Brushless DC motor	III, IV
BVLOS	Beyond visual line of sight	I, V
C2	Command and control link	I, V
CAN	Controller area network bus	IV
CG	Center of gravity	I, IV
CO	Course outcome	All
ConOps	Concept of operations	II, V
DAA	Detect and avoid	I, V
DGCA	Directorate General of Civil Aviation (India)	II, IV, V
DL	Disk loading (T/A)	III
DoD	Depth of discharge	IV
DShot	Digital ESC protocol with checksum	IV
EASA	European Union Aviation Safety Agency	II
EKF	Extended Kalman filter (state estimator)	III, V
EMI / EMC	Electromagnetic interference / compatibility	I, IV
ESC	Electronic speed controller	IV
EVLOS	Extended visual line of sight (with observers)	I
FAA	Federal Aviation Administration (USA)	II
FC	Flight controller	IV
FHSS	Frequency-hopping spread spectrum	I
FOS	Factor of safety	II
FPV	First-person view	III
GCS	Ground control station	I, IV, V
GNSS	Global navigation satellite system (GPS, GLONASS, Galileo, BeiDou, NavIC)	III, V
GRC	Ground risk class (SORA)	V
GSD	Ground sampling distance	III
GSE	Ground support equipment	I

Acronym	Meaning	Unit
HAE	Height above ellipsoid	III
HALE	High-altitude long-endurance	I
HDOP	Horizontal dilution of precision	V
HFE	Heavy-fuel engine	III
HIL / HITL	Hardware-in-the-loop simulation	I, IV
ICAO	International Civil Aviation Organization	III
ICD	Interface control document	I
ICE	Internal combustion engine	III
IMU	Inertial measurement unit	III, IV
I²C	Inter-integrated circuit bus	IV
KV	Motor velocity constant, rpm per volt	IV
L/D	Lift-to-drag ratio	II
LiDAR	Light detection and ranging	III
LiPo	Lithium polymer battery	III, IV
LRE	Launch and recovery equipment	I
LSB	Laminar separation bubble (Unit II); least significant bit (Unit IV)	II, IV
MALE	Medium-altitude long-endurance	I
MAV	Micro air vehicle	III
MAVLink	Micro Air Vehicle Link messaging protocol	IV, V
MEMS	Micro-electro-mechanical systems	I, III
MPCS	Mission planning and control station	I
MSL	Mean sea level	III
MTOM	Maximum take-off mass	I, III
NDVI	Normalized difference vegetation index	III
NIR	Near infrared	III
NTRIP	Networked Transport of RTCM via Internet Protocol	III
OSO	Operational safety objective (SORA)	V
PDB	Power distribution board	IV
PEMFC	Proton-exchange membrane fuel cell	III
PID	Proportional–integral–derivative controller	IV, V
PPK	Post-processed kinematic positioning	III
PWM	Pulse-width modulation	IV
RCS	Radar cross section	II
Re	Reynolds number	II
RINEX	Receiver Independent Exchange format (raw GNSS logs)	III
ROI	Return on investment	III
RPA / RPAS	Remotely piloted aircraft / system	III
RPC	Remote Pilot Certificate (DGCA)	V
RPTO	Remote Pilot Training Organization (DGCA)	V
RTCM	Radio Technical Commission for Maritime Services correction format	III
RTK	Real-time kinematic positioning	III
RTL / RTH	Return to launch / return to home	I, III, V

Acronym	Meaning	Unit
SAIL	Specific assurance and integrity level (SORA)	V
SfM	Structure from motion (photogrammetry)	III
SIL / SITL	Software-in-the-loop simulation	IV, V
SNR	Signal-to-noise ratio	IV
SORA	Specific Operations Risk Assessment	II, V
SPI	Serial peripheral interface	IV
sUAS	Small unmanned aircraft system (≤ 25 kg)	III
T/W	Thrust-to-weight ratio	II, IV
TIR	Thermal infrared	III
UART	Universal asynchronous receiver-transmitter	IV
UAS	Unmanned aircraft system (vehicle + GCS + links + crew + procedures)	I
UAV	Unmanned aerial vehicle (the aircraft only)	I
UIN	Unique Identification Number (Digital Sky)	V
UTM	UAS traffic management	III
VIO	Visual-inertial odometry	III
VLOS	Visual line of sight	I, V
VO	Visual observer	I, V
VRA	Variable-rate application	III
VTOL	Vertical take-off and landing	I, II
W/S	Wing loading	II